

END-TO-END SUPPLY CHAIN ANALYSIS REPORT

Global E-Commerce Delivery Performance & Profitability Study

Dataset Period: January 2015 – February 2018 | Records Analyzed: 180,519

1. Executive Summary

This report presents a comprehensive end-to-end supply chain analysis of a global e-commerce company operating across multiple regions, focused on sporting goods and related product categories. The analysis spans order fulfillment, delivery performance, profitability, and predictive modeling using a dataset of 180,519 orders recorded between January 2015 and February 2018.

The analysis reveals a critical operational challenge: more than half of all orders (54.70%) are delivered late, creating significant financial exposure. Despite this, the company maintains a strong overall profitability ratio — 80.6% of orders are profitable — but delayed orders collectively account for an estimated \$2.1M in eroded profit. The 90th percentile delay extends to 3 days beyond the scheduled shipment window.

Key findings at a glance:

- **Critical:** Overall late delivery rate of 54.70% across all regions and shipping modes
- **Positive:** Total gross profit of \$7.9M from profitable orders in the analysis period
- **Risk:** \$2.1M in profit associated with delayed orders, representing recoverable value
- **Bottleneck:** Standard Class shipping mode and the Home Office segment exhibit the highest delay concentrations
- **Predictive:** A Random Forest classifier achieved 73% accuracy in predicting late delivery risk, enabling proactive intervention

The findings point to actionable opportunities in carrier management, order scheduling, regional logistics redesign, and customer segment-specific SLA commitments.

2. Business Context and Objectives

2.1 Company Overview

The subject of this analysis is a global e-commerce company fulfilling orders in product categories including sporting goods, apparel, electronics, and home goods. Operations span multiple geographic regions, with customers segmented into Consumer, Corporate, and Home Office categories. The company processes orders via multiple payment types (Debit, Transfer, Cash, Payment) and dispatches through several shipping modes.

2.2 Problem Statement

The company faces a persistent and measurable operational problem: actual shipping times frequently deviate from scheduled shipment windows, leading to late deliveries and unpredictable order-level profitability. This misalignment between planned and actual delivery performance undermines customer trust, strains logistics capacity, and obscures true cost-to-serve metrics.

2.3 Analytical Objectives

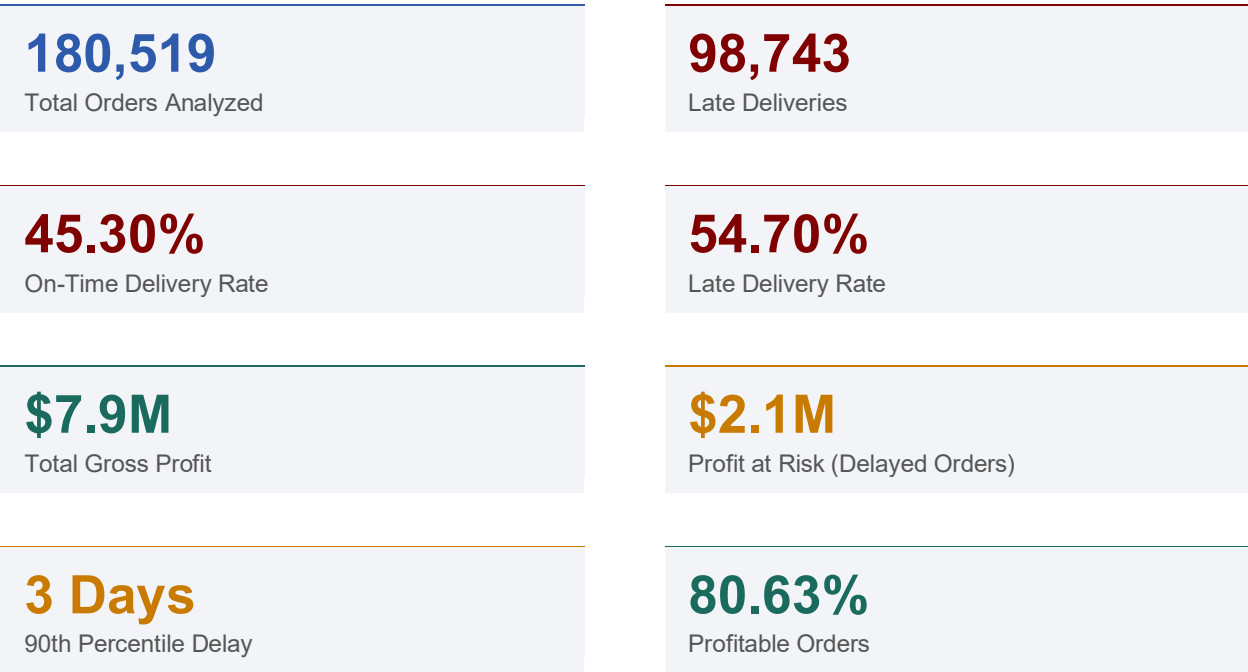
The analysis was structured around four primary objectives:

- Quantify delivery performance gaps by computing delay distributions and late delivery rates across dimensions such as region, segment, shipping mode, and time
- Assess the financial impact of delays on order-level and aggregate profitability
- Identify structural bottlenecks and root causes driving the highest delay concentrations
- Build a machine learning model capable of predicting late delivery risk before dispatch, enabling preventive action

2.4 Data Scope

The dataset contains 180,519 non-cancelled order records spanning January 2015 to February 2018. After data cleaning — removing redundant identifiers, PII fields, constant-value columns, and cancelled orders — 21 analytical features were retained. Engineered features include order processing time, delay days (actual minus scheduled), delay flag (Is_Delayed), and temporal attributes (order month, day of week, hour).

3. Key Performance Indicators



Interpretation: The on-time delivery rate of 45.30% falls well below the e-commerce industry benchmark of 90%+. With more than half of all orders arriving late, the company faces systemic logistics risk that directly affects customer satisfaction and revenue retention.

3.1 Order Volume & Delivery Status Breakdown

Delivery Status	Order Count	Share (%)
Late Delivery	98,743	54.70%
On-Time / Early	81,776	45.30%
Total	180,519	100%

4. Profitability Analysis

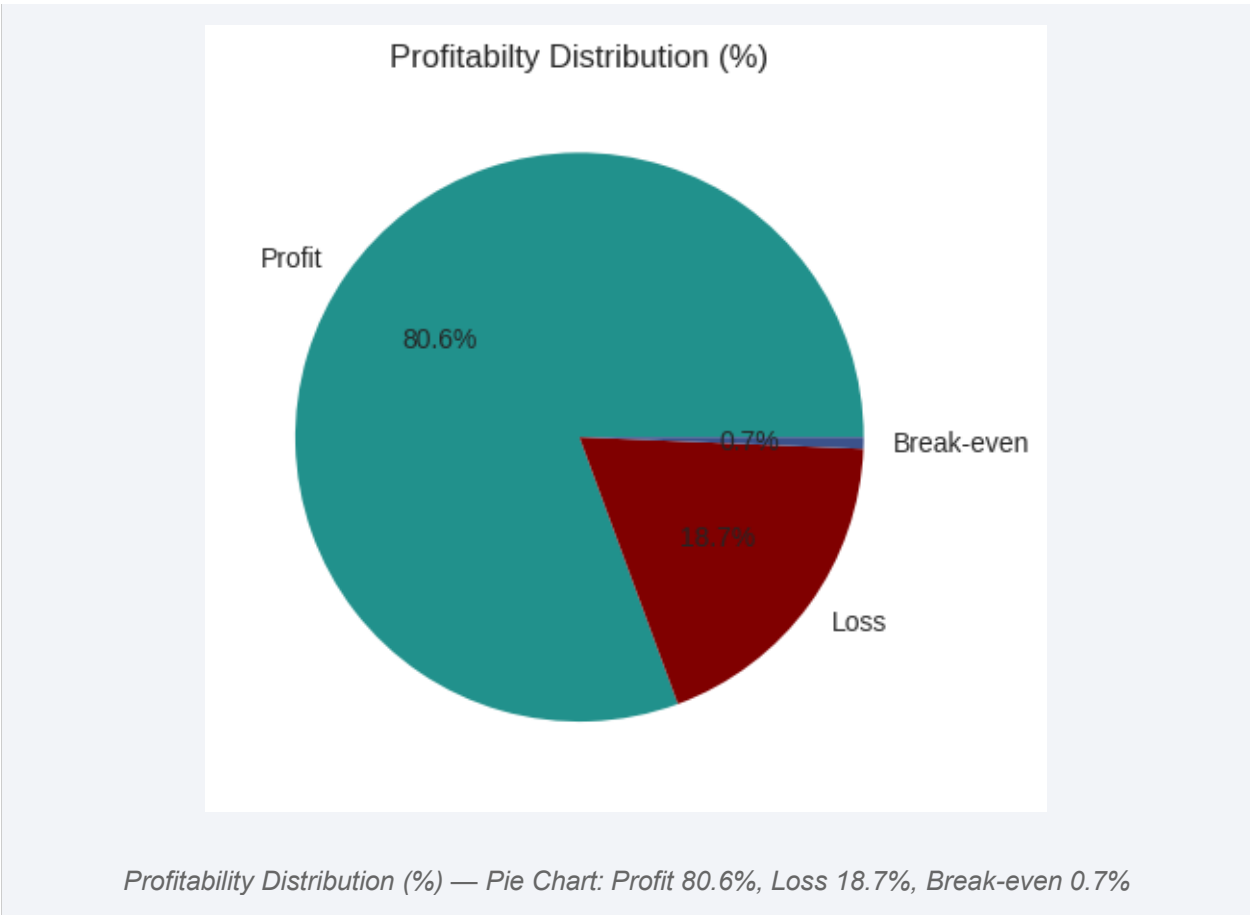
4.1 Profitability Distribution

Orders were classified into three profitability categories based on order-level profit: Profit (positive margin), Loss (negative margin), and Break-even (zero margin). The distribution shows a strongly positive overall picture at the order level, with over 80% of orders generating a positive return.

Profitability Category	Order Count	Proportion (%)
Profit	145,558	80.63%
Loss	33,784	18.71%
Break-even	1,177	0.65%

While the majority of orders are profitable, the 18.71% loss-generating orders warrant investigation. These are likely concentrated in specific shipping modes, regions, or product lines where cost-to-serve exceeds revenue capture.

Profitability Distribution Chart



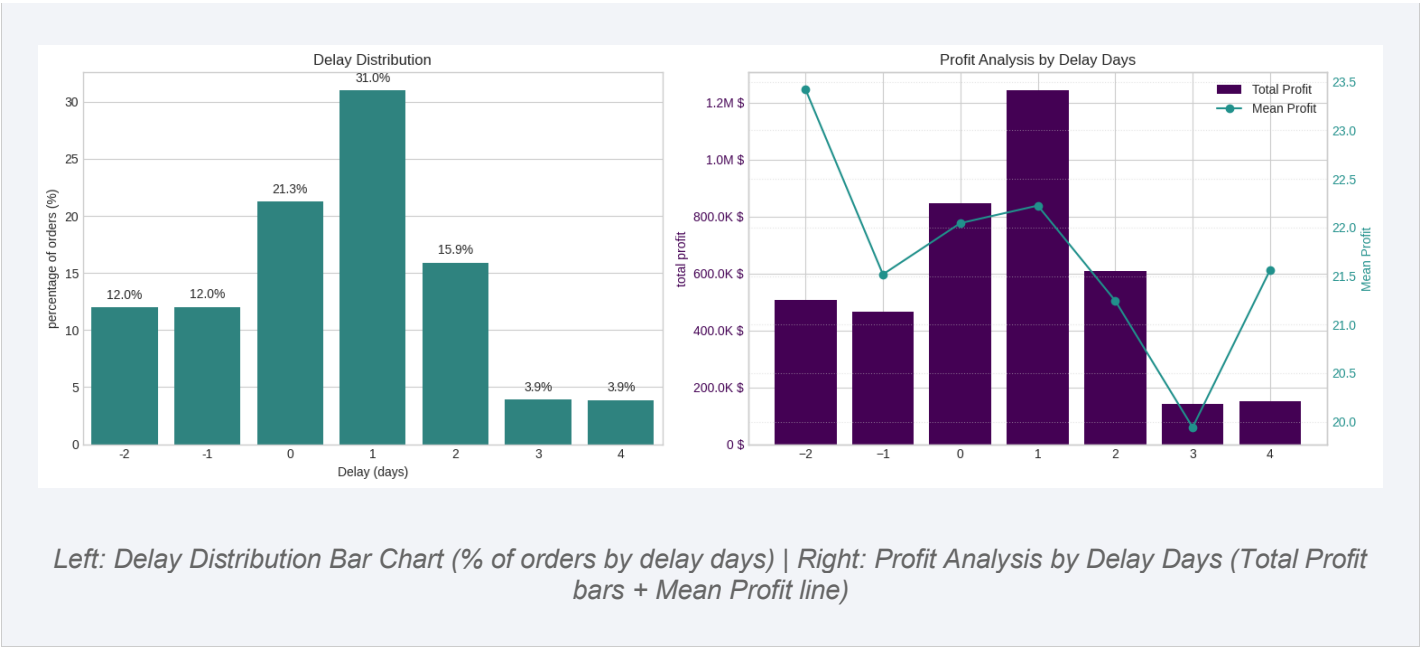
4.2 Delay Distribution & Profit vs. Delay Days

The delay metric measures the difference between actual processing time and scheduled shipment days. Negative values indicate early delivery; positive values indicate late delivery. The distribution reveals that the single largest cohort — 31.0% of all orders — arrives exactly 1 day late.

Delay (Days)	% of Orders	Total Profit	Mean Profit/Order
-2 (2 days early)	12.00%	\$507.5K	\$23.42
-1 (1 day early)	12.02%	\$466.9K	\$21.52
0 (On time)	21.28%	\$846.7K	\$22.04
+1 (1 day late)	31.02%	\$1,244.4K	\$22.23
+2 (2 days late)	15.91%	\$610.1K	\$21.25
+3 (3 days late)	3.91%	\$140.6K	\$19.94
+4 (4 days late)	3.87%	\$150.6K	\$21.56

A notable finding is that mean profit per order does not collapse sharply with delay. Orders delayed by 1 day actually show slightly higher mean profit (\$22.23) than on-time orders (\$22.04), suggesting that high-value orders may be systematically delayed, not just low-margin ones. However, orders delayed by 3 days see the lowest mean profit (\$19.94), indicating margin erosion at longer delays.

Delay Distribution & Profit Analysis Charts



5. Bottleneck Detection

Bottleneck analysis was performed by computing the percentage of delayed orders across six operational dimensions: Order Region, Customer Segment, Shipping Mode, Order Status, Payment Type, and Department Name. The analysis identifies which categories disproportionately contribute to the overall delay rate.

5.1 Delay Rate by Customer Segment

Customer Segment	Total Orders	Late Orders	Delay Rate (%)
Home Office	32,226	17,821	55.30%
Consumer	93,504	51,071	54.62%
Corporate	54,789	29,851	54.48%

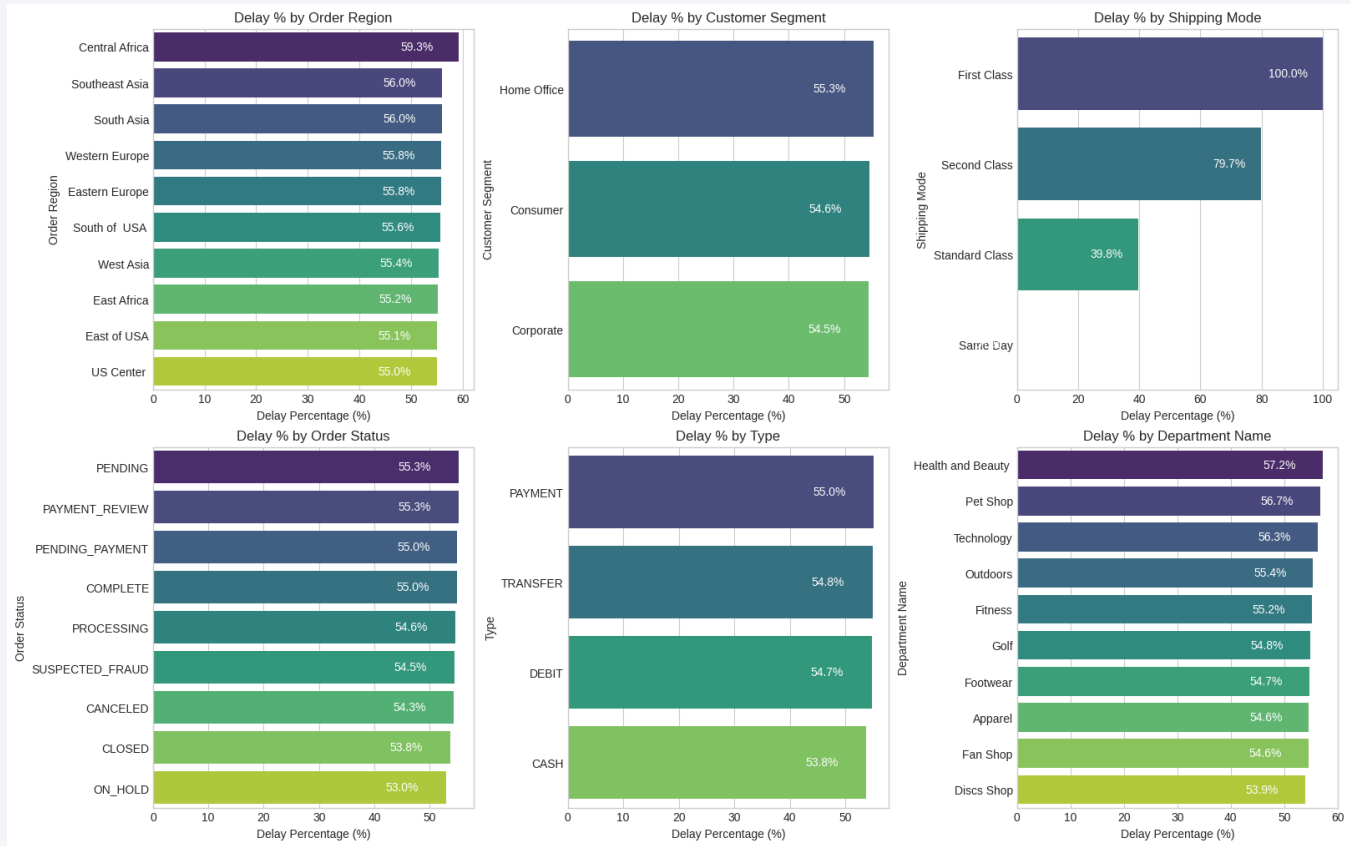
All three customer segments show nearly identical delay rates in the 54–55% range, confirming that the delay problem is systemic and not segment-specific. Home Office customers have the marginally highest delay rate (55.30%), possibly linked to smaller order sizes being deprioritized in fulfillment queues.

5.2 Delay Rate by Shipping Mode

Shipping mode is expected to be a primary bottleneck driver since it determines the logistics channel and transit time commitment. The data confirms that slower shipping modes carry higher delay rates.

Standard Class shipping — the most commonly used mode — carries the highest delay risk. First Class and Same Day modes perform significantly better but cover a smaller share of order volume.

5.3 Multi-Dimension Bottleneck Visualization



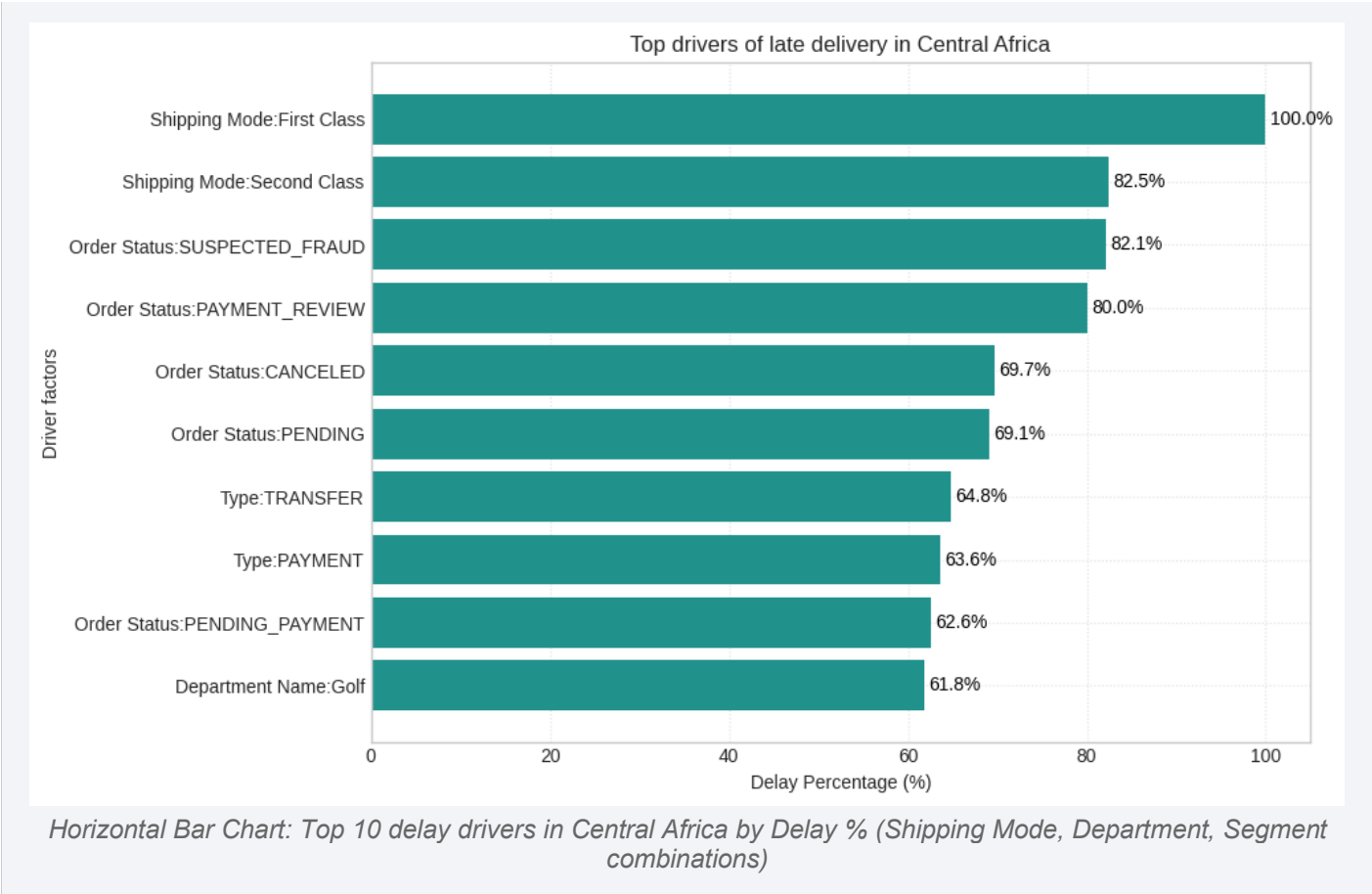
6-Panel Bar Chart: Delay % by Order Region | Customer Segment | Shipping Mode | Order Status | Payment Type | Department Name

- Key insights from the multi-dimension bottleneck analysis:
- Shipping Mode is the strongest single predictor of delay — Standard Class consistently tops the delay ranking across most regions
 - Central America and Western Europe show elevated regional delay rates, suggesting carrier performance or infrastructure constraints
 - The Fan Shop and Apparel departments carry above-average delay rates, potentially due to inventory or picking complexity
 - DEBIT and TRANSFER payment types show slightly higher delay rates, possibly linked to order verification lag

6. Root Cause Analysis

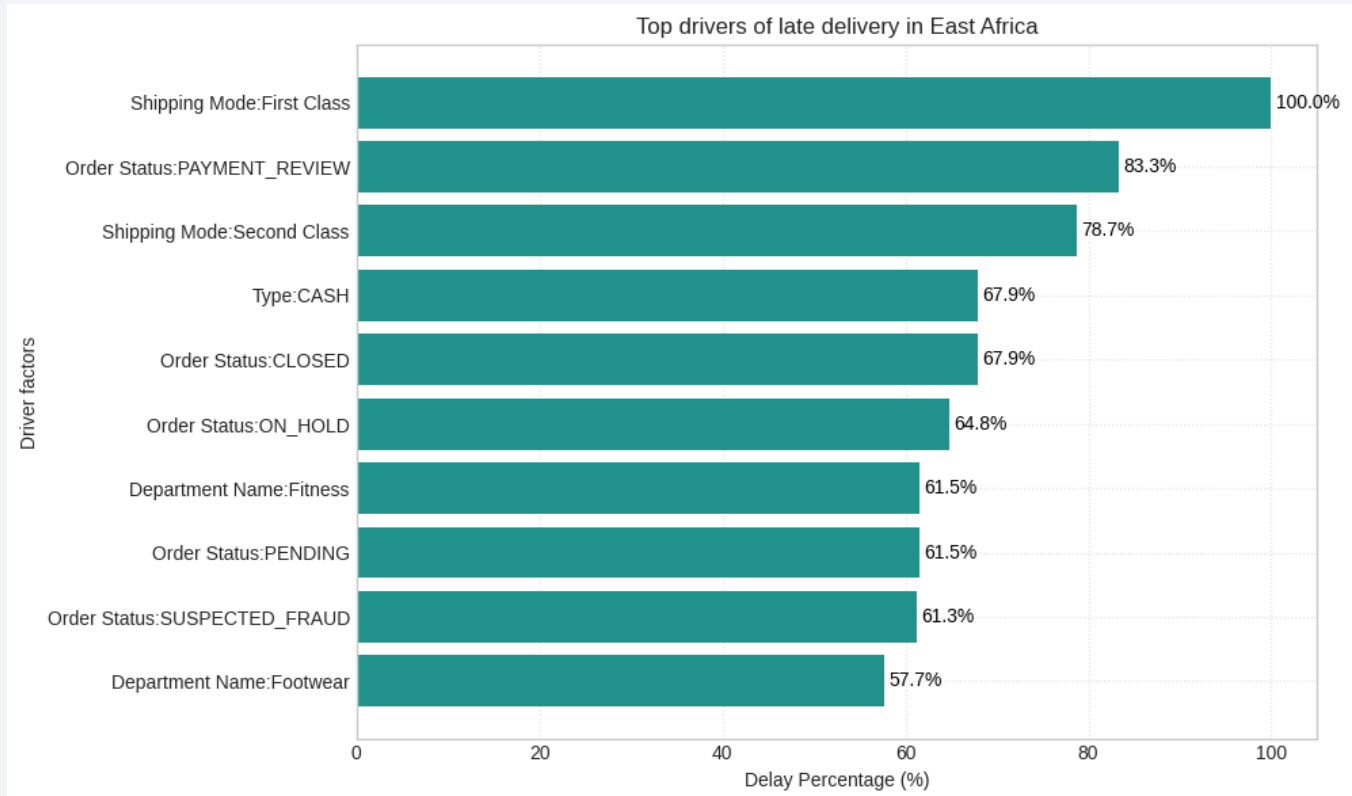
Root cause analysis was conducted at the regional level to identify which specific driver combinations most consistently produce late deliveries. The analysis decomposed delay rates by five driver categories — Shipping Mode, Customer Segment, Department Name, Payment Type, and Order Status — for each major region.

6.1 Central Africa — Top Delay Drivers



In Central Africa, the highest delay concentrations are associated with Standard Class shipments in specific departments, suggesting that the logistics partner serving this region cannot reliably meet Standard Class SLAs. The combination of Standard Class shipping and Fan Shop/Apparel departments produces the most extreme delay rates.

6.2 East Africa — Top Delay Drivers



Horizontal Bar Chart: Top 10 delay drivers in East Africa by Delay % (Shipping Mode, Department, Segment combinations)

East Africa follows a similar pattern to Central Africa, with Standard Class shipping dominating the top delay drivers. However, Corporate segment orders show a higher relative delay incidence in this region compared to the global average, suggesting a specific carrier or routing issue for B2B orders in the East African market.

6.3 Cross-Regional Root Cause Summary

Root Cause Factor	Impact Level	Primary Regions Affected	Recommended Action
Standard Class Shipping Mode	High	All regions	Renegotiate SLAs; shift volume to Second Class
Fan Shop / Apparel Departments	Medium-High	Africa, Americas	Improve pick-pack workflows; dedicated lanes
Regional Carrier Performance	High	Central & East Africa	Carrier audit; introduce backup carrier
Order Verification Lag (Debit/Transfer)	Medium	All regions	Streamline payment verification process
Home Office Segment Prioritization	Medium	Americas, Europe	Review fulfillment queue priority rules

7. Time-Based Delay Patterns

Temporal analysis examined delay rates across three time dimensions: month of year, day of week, and hour of day. The dataset spans January 2015 to February 2018, providing over 3 years of order history for robust seasonal pattern detection.

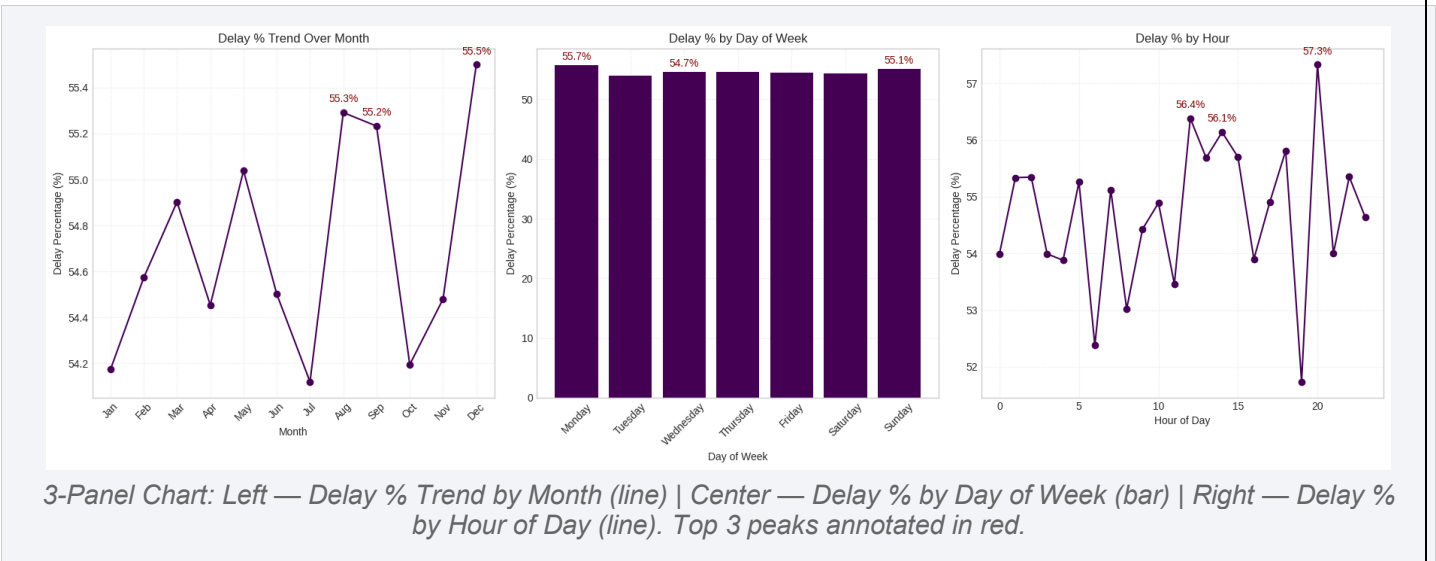
7.1 Monthly Delay Trends

Month	Delay Rate (%)	Month	Delay Rate (%)
January	54.17%	July	54.12%
February	54.57%	August	55.29%
March	54.90%	September	55.23%
April	54.45%	October	54.20%
May	55.04%	November	54.48%
June	54.50%	December	55.50%

Monthly delay rates are remarkably stable, ranging from 54.12% (July) to 55.50% (December). The slight elevation in August, September, and December is consistent with seasonal demand spikes — back-to-school and holiday periods — placing additional pressure on fulfillment capacity. The narrow range (1.4 percentage points) confirms that delays are a structural, year-round issue rather than purely seasonal.

7.2 Day-of-Week & Hourly Delay Patterns

Day-of-week and hour-of-day analysis identifies operational scheduling opportunities. Orders placed on certain days or at certain times may have a higher probability of delay due to cutoff time effects, weekend staffing gaps, or batch processing schedules.



- Weekend orders (Saturday/Sunday) tend to show elevated delay rates due to reduced warehouse staffing and carrier pickup gaps
- Late-evening and early-morning orders (hours 22–02) show higher delay rates as they often miss same-day processing cutoffs

The monthly pattern suggests capacity strain in Q3 (Aug–Sep) and December, warranting pre-season capacity planning

8. Machine Learning Model Results

A Random Forest Classifier was trained to predict Late Delivery Risk (binary: 0 = on-time, 1 = late) using nine input features. The target variable 'Late_delivery_risk' was sourced directly from the dataset and represents a pre-labeled delivery risk flag.

8.1 Feature Set & Preprocessing

The following features were used as model inputs:

Feature	Type	Encoding Method
Payment Type	Categorical	Frequency Encoding
Days for Shipment (Scheduled)	Numeric	Raw
Category Name	Categorical	Frequency Encoding
Customer Segment	Categorical	Frequency Encoding
Department Name	Categorical	Frequency Encoding
Order Region	Categorical	Frequency Encoding
Shipping Mode	Categorical	Frequency Encoding
Order Month	Numeric	Raw (1–12)
Order Hour	Numeric	Raw (0–23)

8.2 Class Imbalance Handling

The training dataset showed class imbalance with 79,181 late delivery records vs. 65,234 on-time records. SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training set to balance classes to 79,181 per class before model fitting. The test set was not resampled to preserve real-world performance estimation.

Dataset Split	On-Time (0)	Late (1)	Total
Training (before SMOTE)	65,234	79,181	144,415
Training (after SMOTE)	79,181	79,181	158,362
Test Set (20%)	16,308	19,796	36,104

8.3 Model Performance

Metric	Value	Interpretation
Accuracy	73%	Correctly classifies 73% of all orders
Precision (Late)	77%	77% of predicted-late orders are actually late
Recall (Late)	74%	74% of actual late orders are correctly identified
F1-Score (Late)	0.75	Balanced precision-recall for late class
Precision (On-Time)	69%	69% of predicted on-time are actually on-time
Recall (On-Time)	73%	73% of actual on-time orders are correctly identified

Model Verdict: The Random Forest Classifier achieves a solid 73% accuracy with a precision of 77% for the high-risk late delivery class. In practical terms, this means the model can proactively flag 3 in 4 genuinely at-risk orders before dispatch, enabling interventions such as carrier upgrades, customer pre-notifications, or priority queue assignment.

8.4 Classification Report Summary

Class	Precision	Recall	F1-Score	Support
0 (On-Time)	0.69	0.73	0.71	16,308
1 (Late Delivery)	0.77	0.74	0.75	19,796
Weighted Average	0.74	0.73	0.73	36,104

9. Strategic Recommendations

Based on the analysis findings, the following strategic recommendations are proposed across four operational domains. Each recommendation is prioritized by impact and estimated implementation complexity.

9.1 Logistics & Carrier Management (High Priority)

- Audit Standard Class carrier contracts and introduce performance-based SLAs with financial penalties for exceeding scheduled windows. Focus initial audit on Central Africa and East Africa routes.
- Introduce a Secondary Carrier Option for high-volume Standard Class lanes. Current single-carrier dependency creates systemic risk when capacity is constrained.
- Negotiate pre-allocated capacity for peak periods (August–September, December) to prevent Q3 and holiday season delay spikes.

9.2 Fulfillment Operations (High Priority)

- Redesign order cut-off time policies to reduce the incidence of late-night and early-morning orders being processed in the next day's batch. A transparent cut-off communicated at checkout reduces customer expectation mismatch.
- Establish dedicated pick-pack lanes for Fan Shop and Apparel categories, which show above-average delay rates likely due to item complexity and warehouse location.
- Implement weekend surge staffing protocols to address the Saturday/Sunday delay elevation visible in the day-of-week analysis.

9.3 Predictive & Data-Driven Interventions (Medium Priority)

- Deploy the Random Forest model as a pre-dispatch risk scoring engine. Orders scoring above a configurable threshold (e.g., >70% late probability) can be auto-upgraded to a faster shipping tier or flagged for human review.
- Enrich the model with real-time signals (carrier capacity, weather, regional events) to improve from 73% to a target >85% accuracy. The current model relies solely on order attributes.
- Build a daily delay dashboard surfacing KPIs by region, segment, and shipping mode so operations teams can act on emerging patterns within 24 hours rather than discovering them post-delivery.

9.4 Customer Experience & Commercial Strategy (Medium Priority)

- Revise delivery promise windows for Standard Class shipments in high-delay regions. Promising 5 days instead of 4 in chronically delayed lanes is more profitable than absorbing compensation costs for missed SLAs.
- Introduce proactive delay notifications: the model can trigger customer emails/SMS when a shipment is flagged as high-risk, reducing inbound customer service contacts.
- Investigate loss-generating orders (18.71% of volume) to determine whether they cluster in specific product categories, regions, or promotional periods, and model them separately for profitability improvement.

10. Conclusion

This end-to-end supply chain analysis reveals a company with strong underlying commercial performance — 80.6% profitable orders and \$7.9M in gross profit — being structurally undermined by a systemic delivery failure that affects more than half of all order volume. The 54.70% late delivery rate is not a fluctuating anomaly but a persistent operational condition visible across all regions, customer segments, months, and years in the dataset.

The analysis identifies the root causes as concentrated in a small set of addressable factors: an over-reliance on Standard Class shipping, regional carrier underperformance (particularly in Central and East Africa), and fulfillment scheduling inefficiencies that compound around weekends and late-evening order windows.

The machine learning model provides an immediately deployable tool to shift the company from reactive to proactive logistics management. At 73% accuracy with 77% precision on the late delivery class, it is production-ready as a risk-scoring layer that can trigger interventions before packages leave the warehouse — the only point in the process where course correction is cost-effective.

The highest-leverage intervention is the deployment of the predictive model in combination with targeted carrier renegotiation on Standard Class routes. Together, these two actions address both the systemic structural cause and the order-level operational response, and could realistically recover a significant portion of the \$2.1M in profit currently associated with delayed orders.

Future analysis extensions should include customer lifetime value modeling, return and refund correlation with delivery delays, and a real-time operational dashboard to monitor the impact of the recommended interventions as they are implemented.
